

Maths1
3rd guided works

Exercise 1

Prove by induction that :

- 1) $\forall n \in \mathbb{N} : 10^n - (-1)^n$ is multiple of 11.
- 2) $\forall n \in \mathbb{N} - \{3\} : 2^n \geq 2$.
- 3) $\forall n \in \mathbb{N} : 1 + x + x^2 + \dots + \frac{1 - x^{n+1}}{1 - x} \quad x \in \mathbb{R} - \{1\}$

Exercise 2

For each of the following subsets of $\mathbb{R}$, determine if it is upper bounded, lower bounded, and bounded. If yes, provide its supremum, and/ or infimum.

- 1- $A = \{x \in \mathbb{R} / 0 < x < \sqrt{3}\}$.
- 2- $B = \{x \in \mathbb{R} / \frac{1}{2} \leq \sin x < \frac{\sqrt{3}}{2}\}$
- 3- $C = \{x \in \mathbb{R} / x^3 > 3\}$.
- 4- $D = \{x \in \mathbb{R} / e^x < \frac{1}{2}\}$

Exercise 3

Let $A = [2, 4[\cup]0, 1]$, $B = \{x^2 + 1/x \in [1, 2]\}$, $C = \{\frac{1}{n}/n \in \mathbb{N}^*\}$, $D = \{n + 1/n \in \mathbb{N}^*\}$

- 1- Are the sets A, B, C and D bounded sets.
- 2- Calculate their supremum and infimum and find their maximum and minimum if they exist.

Exercise 4

Determine the domain of definition of each of the following functions.

$$f(x) = \sqrt{x^2 - 3x + 2} + \sqrt{2x - 3}$$
$$g(x) = \frac{\ln(x-3)}{f(x)}, \quad h(x) = \sqrt{4 - 3x^2}$$

Exercise 5

Let f, g and h be three functions defined by :

$$f(x) = \begin{cases} \frac{\sqrt{x}-1}{\sqrt[3]{x}-1} & \text{if } x \neq 1 \\ 2\alpha & \text{if } x = 1 \end{cases}, \quad g(x) = \begin{cases} \frac{1 - \sqrt{\cos x}}{x^2} & \text{if } x \neq 0 \\ 3\beta - 1 & \text{if } x = 0 \end{cases}, \quad h(x) = \begin{cases} \frac{x^2 - |x|}{x^2 + |x|} & \text{if } x \neq 0 \\ 3\gamma - 2 & \text{if } x = 0 \end{cases}$$

For what values of α, β, γ are the functions f, g and h continuous on $\mathbb{R}$.

Exercise 6

- 1- Show that there exists $c \in [0, 1]$ such that $\ln(1 + c^2) = 1 - c$
- 2- Is c unique.

Exercise 7

Solve the following equations :

1- $\cos(\arcsin x) = \frac{1}{\sqrt{2}}$

2- $\sin(\arccos x) = \frac{\sqrt{3}}{2}$

3- $\arcsin 2x - \arcsin \sqrt{3}x = \arcsin x$.

Exercise 8

Let a and b be two positive real numbers such that $a^2 - b^2 = 1$.

Solve the system of equations :

$$\begin{cases} \cosh x + \cosh y = 2a \\ \sinh x + \sinh y = 2b \end{cases}$$